

Daniel Olofin - Interest in Statistical Methodology

My motivation to pursue statistical methodology stems from both personal experience and scientific curiosity. Growing up within a medically underserved community and later navigating a prolonged illness during college where several treatments failed before one succeeded. I witnessed firsthand how uncertainty shapes clinical decisions. That experience sparked a lasting question that continues to guide my work: **how can rigorous, interpretable, and scalable statistical methods reduce uncertainty and improve therapeutic decision-making?**

This motivation shaped my early research trajectory. My first independent project, modeling HIV prevalence in Nigeria using ARIMA, taught me the fundamentals of time-series inference and the importance of aligning statistical assumptions with real-world biological processes. To address the limitations, I observed in ARIMA particularly its inability to capture nonlinear dynamics. I collaborated with DeYoung Data Analytics on a comparative analysis of ARIMA, ARFIMA, RNNs, and ANNs. This expanded my appreciation for the trade-offs between classical statistical modeling and machine learning, especially interpretability versus predictive power.

My most formative research experience occurred at the **Clonal Redesign Lab at Moffitt Cancer Center**, where I made the transition from prediction-focused modeling to **mechanistic statistical modeling in oncology**. There, I developed a discrete-time Markov chain model to examine how chromosomal instability and karyotype evolution respond to metabolic stress in cancer cells. By grounding transition probabilities in biological missegregation mechanisms, I demonstrated that control populations follow neutral drift (with simulations matching empirical late-phase karyotypes at $p = 1 \times 10^{-5}$), while glucose-deprived populations show strong divergent trajectories indicative of selection. This project reshaped my understanding of statistical methodology, not merely as a descriptive tool, but as a framework capable of uncovering underlying biological mechanisms and informing therapeutic hypotheses.

My methodological interests further expanded through my role as a **Junior Statistician at the Center for Clinical Trials and Research Implementation Science (CCTRIS)**, where I supported randomization workflows, data monitoring, and statistical coordination for ongoing clinical studies. Working closely with clinical research teams exposed me to the operational and methodological challenges underlying trial design, including missing data, interim monitoring decisions, and reproducibility concerns. These experiences solidified my interest in developing statistical methods that are both theoretically grounded and operationally impactful.

A complementary dimension of my methodological development emerged through my creation of **SmartMatch**, an adaptive R Shiny application designed to automate real-time patient eligibility screening in clinical trials. Motivated by the reality that **nearly 80% of clinical trials struggle to recruit sufficient participants**, I built SmartMatch to integrate database systems with statistical screening logic, streamlining workflows and improving equity in patient identification. Its translational value was recognized through its acceptance for presentation at the **2025 Boston Pharmaceutical Statistics Symposium**. This project strengthened my skills in statistical computing, software engineering, and clinical informatics, and reinforced my belief that methodological innovation should be paired with real-world applicability.

Together, these experiences (my personal motivation, methodological research, clinical trial involvement, and software development) have shaped my desire to pursue advanced training in **adaptive platform trial methodology**. I am eager to contribute to developing statistical tools that improve efficiency, interpretability, and decision-making in modern trial designs.